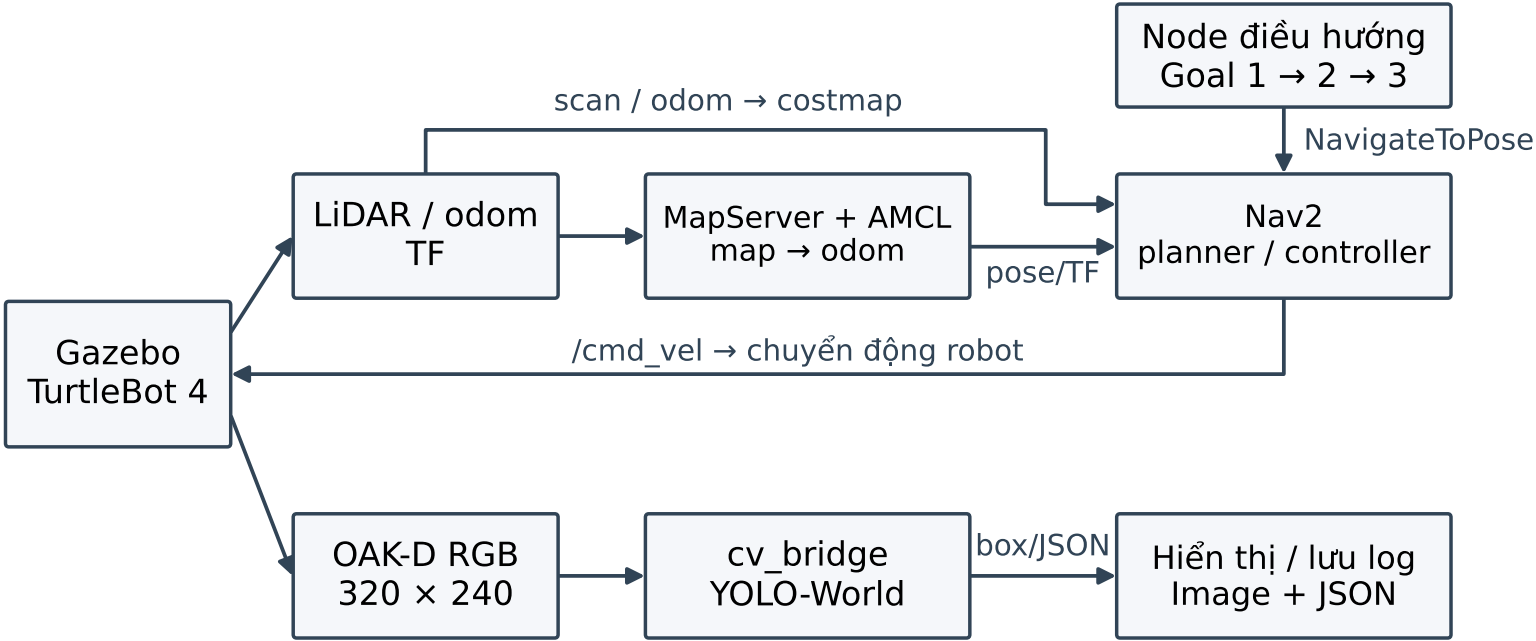
Node điều hướng
Goal 1 → 2 → 3
NavigateToPose
scan / odom → costmap
LiDAR / odom
TF
MapServer + AMCL
map → odom
Nav2
planner / controller
pose/TF
Gazebo
TurtleBot 4
/cmd_vel → chuyển động robot
OAK-D RGB
320 × 240
cv_bridge
YOLO-World
box/JSON
Hiển thị / lưu log
Image + JSON